GATE
CS & DA

Artificial Intelligence

Proposition & Predictive Logic

Lecture No. 10

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Recap of Previous Lecture

Topic

Topic

Topic

Topic

Topic

Predicate

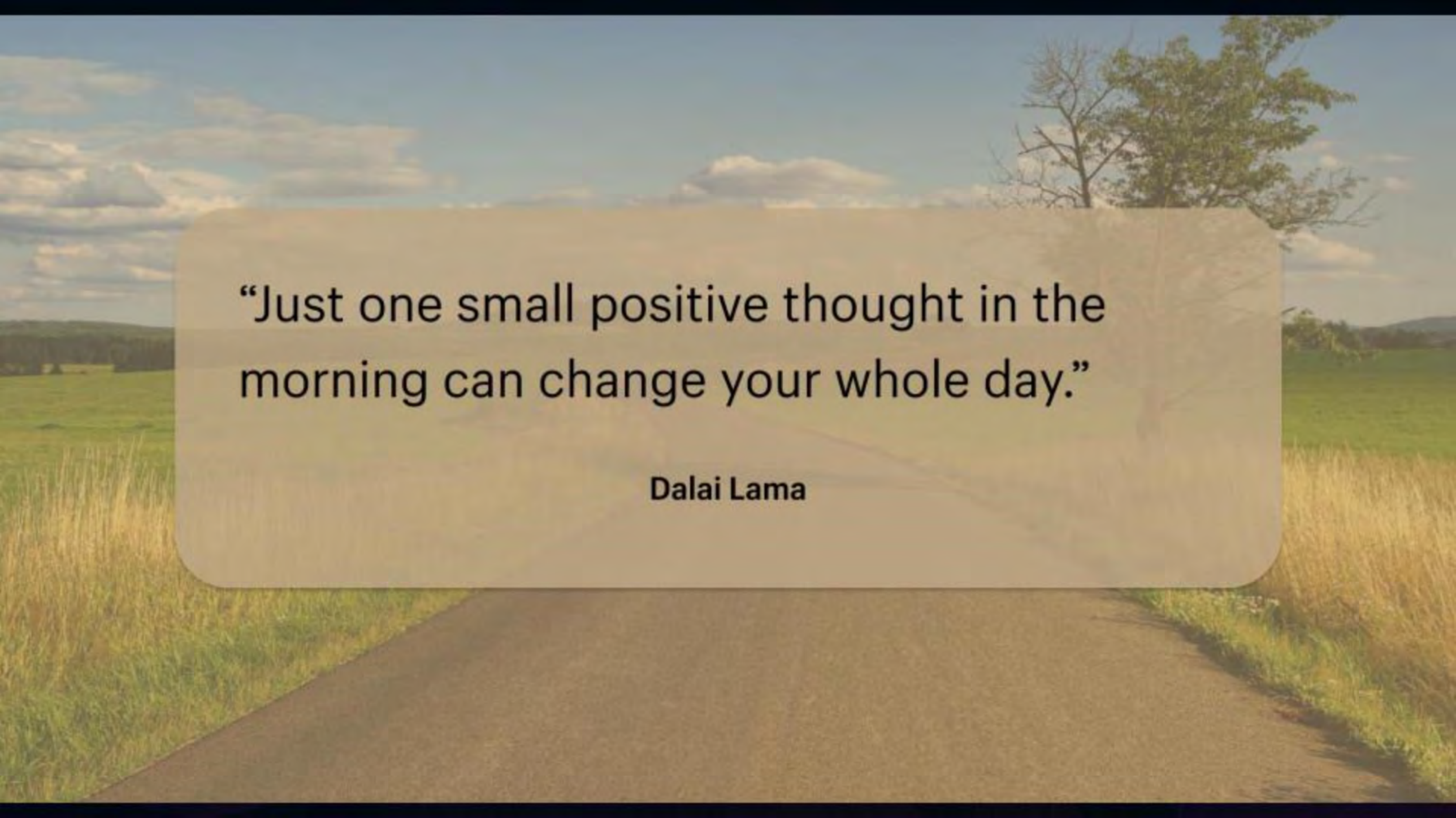
+

Propositional Logic

Turn on Slide map

Topics to be Covered

- Topic Questions
- Topic
- Topic
- Topic
- Topic


“Just one small positive thought in the morning can change your whole day.”
Dalai Lama

[NAT]

5. $\forall x\,[P \to Q(x)] \equiv P \to \forall x\, Q\, x$

Yes

6. $\exists x\,[P \to Q(x)] \equiv P \to \exists x\, Q(x)$

$\exists x\,(\sim P \vee Q(x)$

7. $\forall x\,[P(x) \to Q] \equiv \exists x\, P(x) \to Q$

$\forall x\,[\sim P(x) \vee Q]$

$(\forall x \sim P(x) \vee Q)$

$\sim(\sim\forall x \sim P(x) \vee Q)$

$\sim(\exists x P(x) \vee Q$

8. $\exists x\,[P(x) \to Q] \equiv \sim \forall x\, P(x) \to Q$

[MCQ]

Which of the following predicate calculus statements is/are valid.

If this is true (Yes)

A. $(\forall x)\, P(x) \vee (\forall x)\, Q(x) \rightarrow (\forall x)\{P(x) \vee Q(x)\}$

B. $(\exists x)\, P(x) \wedge (\exists x)\, Q(x) \rightarrow (\exists x)\{P(x) \wedge Q(x)\}$ ← False

C. $(\exists x)\, \{P(x) \vee Q(x)\} \rightarrow (\forall x)\, P(x) \vee (\forall x)\, Q(x)$

D. $(\exists x)\, \{P(x) \vee Q(x)\} \rightarrow \{\exists x\, P(x) \vee \forall x\, Q(x)\}$

P	Q
0	1
0	0
0	1

[MCQ]

Which of the following predicate calculus statements is/are valid

$\forall x\,(P(x) \to Q(x)) \Rightarrow \forall x\,P(x) \to \forall x\,Q(x)$

$\forall x\,P(x) \to \forall x\,Q(x) \Rightarrow \forall x\,(P(x) \to Q(x))$

(A) $\forall(x)\,(P(x) \Rightarrow Q(x)) \Rightarrow ((\forall(x)\,P(x)) \Rightarrow (\forall(x)\,Q(x)))$

(B) $\exists x(P(x) \vee Q(x)) \Rightarrow ((\exists x P(x)) \Rightarrow (\exists x Q(x)))$

P	Q
1	0
0	0
1	0

$1 \to 0$

(C) $\exists x(P(x) \wedge Q(x)) \Leftrightarrow ((\exists x P(x)) \wedge (\exists x Q(x)))$

$\exists x\,(P(x) \wedge Q(x)) \to (\exists x\,(P(x)) \wedge \exists x\,Q(x))$

P	Q
1	0
0	1

(D) $\forall(x)\,\exists y\,P(x, y) \Rightarrow \exists y\,\forall(x)P(x, y)$

[MCQ]

Let P(x) and Q(x) be arbitrary predicates. Which of the following statements is always TRUE?

P	Q	P∨Q
1	0	1
0	1	1

(A) $(\forall x(P(x) \vee Q(x))) \Rightarrow ((\forall x P(x)) \vee (\forall x Q(x)))$

(0 ∨ 0)

P	Q
1	0
0	1

(B) $(\forall x(P(x) \Rightarrow Q(x))) \Rightarrow ((\forall x P(x)) \Rightarrow (\forall x Q(x)))$ True

(C) $(\forall x\, (P(x) \Rightarrow \forall\, x\, Q(x))) \Rightarrow (\forall x\, (P(x) \Rightarrow Q(x)))$

(D) $((\forall x(P(x)) \Leftrightarrow (\forall x\, Q(x))) \Rightarrow (\forall x(P(x) \Leftrightarrow Q(x)))$

(0 ⇔ 0) ✓ false

True

$\forall x(P(x) \equiv Q(x)) \Rightarrow \forall x(P(x)) \equiv \forall x(Q(x))$

P	Q	PQ	PQ
1	1	0	1
1	1	0	0
1	1	0	0
			1

[MCQ]

Which of the following is a valid first order formula? (Here α and β are first order formulae with x as their only free variable)

A. False $((\forall x)[\alpha] \Rightarrow (\forall x)[\beta]) \Rightarrow (\forall x)\,[\alpha \Rightarrow \beta]$

B. True False $((\forall x)[\alpha]) \Rightarrow (\exists x)[\alpha \wedge \beta])$

C. $(\forall x)[\alpha \vee \beta] \Rightarrow ((\exists x)[\alpha] \Rightarrow (\forall x)[\beta])$

D. $((\forall x)[\alpha \Rightarrow \beta] \Rightarrow ((\forall x)[\alpha]) \Rightarrow (\forall x)[\beta])$

α	β	α ∧ β
1	0	0
1	0	0
1	0	0
1	0	0

α	β
1	0
0	1

$\exists x(\alpha) \Rightarrow \forall x \beta$

$1 \Rightarrow 0$

$\forall x\,[\alpha \vee \beta] \Rightarrow (\exists x(\alpha) \Rightarrow \exists x(\beta))$ False

α	β
1	0
1	0

$1 \Rightarrow 0$

$\forall x \alpha \Rightarrow 1$

α $\quad \exists x \alpha \Rightarrow$ α shd be 1 at least one time $\Rightarrow 1$

[MCQ]

Which of the following first order formulae is logically valid? Here $\alpha(x)$ is a first-order formula with x as a free variable and β is a first-order formula with no free variable.

A. $[\beta \to (\exists x, \alpha(x))] \to [\forall x, \beta \to \alpha(x)]$ $= (\beta \to \forall x \alpha(x))$

B. $[\exists x, \beta \to (\alpha(x))] \to [\beta \to (\forall x, \alpha(x))]$

C. $[\exists x, \alpha(x)) \to \beta] \to [\forall x, \alpha(x) \to \beta]$

LHS True

D. $[\forall x, \alpha(x)) \to \beta] \to [\forall x, \alpha(x) \to \beta]$

β: False

1) LHS/RHS True

2) β: True

$\exists x(\alpha) =$ True

LHS: True

$\forall x(\alpha) =$ False

$\beta \to \forall x(\alpha)$

True $\to$ False

RHS False

• β True

LHS, RHS $\to$ True

β False

• $\exists x(\alpha) = 0$

This means $\alpha = 0$ for all x

Under this

RHS: True

Q. Choose among the following that are not equivalent to the given first order logic statement:

• ABC are not equivalent

$(\exists x)(\forall y)[p(x, y) \wedge q(x, y) \wedge \neg r(x, y)]$

(A) $(\forall x)(\exists y)[p(x, y) \wedge q(x, y) \rightarrow r(x, y)]$

(B) $(\exists x)(\forall y)[p(x, y) \vee q(x, y) \wedge \neg r(x, y)]$

(C) $\neg(\forall x)(\exists y)[p(x, y) \vee q(x, y) \rightarrow r(x, y)]$

(D) $\neg(\forall x)(\exists y)[p(x, y) \wedge q(x, y) \rightarrow r(x, y)]$

$(\neg A \vee B) \equiv (A \rightarrow B)$

$\sim(\sim P \vee \sim Q) \Rightarrow (P \wedge Q)$

$\exists x \forall y($

$\sim\sim \exists x \forall y (P \wedge Q \wedge \neg R)$

$\sim \forall x \exists y (\sim P \vee \sim Q \vee R)$

$\sim \forall x \exists y ((P \wedge Q) \rightarrow R)$

Which of the following is not valid logical expression?

(A) $\forall x\,[P(x) \rightarrow Q(x)] \rightarrow [\forall x\, P(x)] \rightarrow [\forall x\, Q(x)]$

(B) $\forall x\,[P(x) \vee Q(x)] \rightarrow [\forall x\, P(x)] \vee [\forall x\, Q(x)]$

(C) $\exists x\,[P(x) \wedge Q(x)] \rightarrow [\exists x\, P(x)] \wedge [\exists x\, Q(x)]$

(D) $\forall x\,[P(x) \leftrightarrow Q(x)] \rightarrow [\forall x\, P(x)] \leftrightarrow [\forall x\, Q(x)]$

Consider following two First Order Logic Statements:

S_1: $[\forall x(\sim P(x) \vee Q(x))] \rightarrow [\forall x\, P(x)] \rightarrow [\forall x\, Q(x)]$

S_2: $[\exists x\, P(x)] \rightarrow [\exists x\, Q(x)] \rightarrow [\exists x\, (P(x) \rightarrow Q(x))]$

Which of the following is valid?

(A) S_1 only

(B) S_2 only

(C) Both S_1 and S_2

(D) None of these

$\forall x(\sim P \vee Q) \equiv \forall x(P \rightarrow Q)$

$\downarrow$

$\forall x P \rightarrow \forall x Q$

• LHS = True

$\exists_x P$	$\exists_x Q$
0	0
0	1
1	0
1	1

$\Rightarrow$ LHS, RHS True

P	Q	
1	0	F
0	1	T
1	1	T
0	0	T
0	1	T

$P(y) = \sqrt{y}$ is real in the domain of Z^+ then which of the following is / are correct?

(A) $\forall$ P(y)

(B) $\exists$y P(y)

(C) $\forall$y ~ P(y)

(D) $\exists$y ~ P(y)

Let p and q be the following propositions:

p: Fail grade can be given.

q: Student scores more than 50% marks

Consider the statement "Fail grade cannot be given when student scores more than 50% marks"

Which one of the following is the CORRECT representation of the above statement in propositional logic?

(a) $q \rightarrow \neg p$ (b) $q \rightarrow p$

(c) $\neg p \rightarrow q$ (d) $p \rightarrow q$

more than 50% marks → fail grade cannot be given

The statement $\boxed{(\neg p) \Rightarrow (\neg q)}$ is logically equivalent to which of the statements below?

Contra positive.

$\neg P \rightarrow \neg Q$

I. $p \Rightarrow q$

II. $q \Rightarrow p$

III. $(\neg q) \vee p$

IV. $(\neg p) \vee q$

(a) I only

(b) I and IV only

(c) II only

(d) II and III only

Consider the following two statements:

S_1: If a candidate is known to be corrupt, then he will not be elected. $\neg q$ p

S_2: If a candidate is kind, he will be elected. R

Which one of the following statements follows from S_1 and S_2 as per sound inference rules of logic?

(a) If a person is known to be corrupt, he is kind

(b) If a person is not known to be corrupt, he is not kind

(c) If a person is kind, he is not known to be corrupt

(d) If a person is not kind, he is not known to be corrupt

P: Corrupt
Q: elected
R: kind

- $P \to \neg Q$
- $R \to Q$
- $\neg Q \to \neg R$
- $Q \to \neg P$
- $\boxed{P \to \neg R}$
- $R \to \neg P$

Consider the following statements:

P: Good mobile phones are not cheap

Q: Cheap mobile phones are not good

L: P implies Q

M: Q implies P

N: P is equivalent to Q

H.W

Which one of the following about L, M, and N is CORRECT?

(a) Only L is TRUE

(b) Only M is TRUE

(c) Only N is TRUE

(d) L, M and N are TRUE

Which one of the following is NOT logically equivalent to $\neg\exists x(\forall y(\alpha) \wedge \forall z(\beta))$?

(a) $\forall x(\exists z(\neg\beta) \rightarrow \forall y(\alpha))$ H.W

(b) $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha))$

(c) $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg\beta))$

(d) $\forall x(\exists y(\neg\alpha) \vee \exists z(\neg\beta))$

Which one of the following predicate formulae is NOT logically valid?

Note that W is a predicate formula without any free occurrence of x.

H.W

(a) $\exists x(p(x) \wedge W) \equiv \exists x\, p(x) \wedge W$

(b) $\exists x(p(x) \rightarrow W) \equiv \forall x\, p(x) \rightarrow W$

(c) $\forall x\, (p(x) \rightarrow W) \equiv \forall x\, p(x) \rightarrow W$

(d) $\forall x\, (p(x) \vee W) \equiv \forall x\, p(x) \vee W$

Consider the first-order logic sentence.

F: $\forall x(\exists y R(x, y))$. Assuming non-empty logical domains, which of the sentences below are implied by F?

H.W

I. $\exists y(\exists x\ R(x, y))$

II. $\exists y(\forall x\ R(x, y))$

III. $\forall y(\exists x R(x, y))$

IV. $\neg\exists x(\forall y \neg R(x, y))$

(a) IV only

(b) I and IV only

(c) II only

(d) II and III only

Which one of the following well-formed formulae in predicate calculus is NOT valid? H.W

(a) $(\forall x\, p(x) \Rightarrow \forall x\, q(x)) \Rightarrow (\exists x\, \neg p(x) \vee \forall x\, q(x))$

(b) $(\exists x\, p(x) \vee \exists x\, q(x)) \Rightarrow \exists x(p(x) \vee q(x))$

(c) $\exists x(p(x) \wedge q(x)) \Rightarrow (\exists x\, p(x) \wedge \exists x\, q(x))$

(d) $\forall x(p(x) \vee q(x)) \Rightarrow (\forall x\, p(x) \vee \forall x\, q(x))$

Which of the following is the negation of

$[\forall x, \alpha \rightarrow (\exists y, \beta \rightarrow (\forall u, \exists v, \gamma))]$? H.W

(a) $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \gamma))]$

(b) $[\exists x, \alpha \rightarrow (\forall y, \beta \rightarrow (\exists u, \forall v, \neg \gamma))]$

(c) $[\forall x, \neg \alpha \rightarrow (\exists y, \neg \beta \rightarrow (\forall u, \exists v, \neg\gamma))]$

(d) $[\exists x, \alpha \wedge (\forall y, \beta \wedge (\exists u, \forall v, \neg\gamma))]$

Consider the following well-formed formulae:

I. $\neg\forall x(P(x))$ II. $\neg\exists x(P(x))$

III. $\neg\exists x(\neg P(x))$ IV. $\exists x(\neg P(x))$

Which of the above are equivalent?

H.W

(a) I and III (b) I and IV

(c) II and III (d) II and IV

Which one of the following is NOT logically equivalent to $\neg\exists x(\forall y(\alpha) \wedge \forall z(\beta))$?

(a) $\forall x(\exists z(\neg\beta) \rightarrow \forall y(\alpha))$ H.W

(b) $\forall x(\forall z(\beta) \rightarrow \exists y(\neg\alpha))$

(c) $\forall x(\forall y(\alpha) \rightarrow \exists z(\neg\beta))$

(d) $\forall x(\exists y(\neg\alpha) \vee \exists z(\neg\beta))$

Geetha has a conjecture about integers, which is of the form

$\forall x(P(x) \Rightarrow \exists y Q(x, y)$,

where P is a statement about integers, and Q is a statement about pairs of integers. Which of the following (one or more) option(s) would imply Geetha's conjecture?

H.W

(a) $\exists x(P(x) \wedge \forall y Q(x, y))$

(b) $\forall x \forall y Q(x, y))$

(c) $\exists y \forall y (P(x) \Rightarrow Q(x, y))$

(d) $\exists x(P(x) \wedge \exists y Q(x, y))$

Which one of the following is Not equivalent to $p \leftrightarrow q$?

(a) $(\neg p \vee q) \wedge (p \vee \neg q)$

(b) $(\neg p \vee q) \wedge (q \rightarrow p)$ H.W

(c) $(\neg p \wedge q) \vee (p \wedge \neg q)$

(d) $(\neg p \wedge \neg q) \vee (p \wedge q)$

Consider the following two statements:

S_1: If a candidate is known to be corrupt, then he will not be elected.

S_2: If a candidate is kind, he will be elected.

Which one of the following statements follows from S_1 and S_2 as per sound inference rules of logic?

(a) If a person is known to be corrupt, he is kind

(b) If a person is not known to be corrupt, he is not kind

H.W

(c) If a person is kind, he is not known to be corrupt

(d) If a person is not kind, he is not known to be corrupt

THANK - YOU